

João José

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joaotiagojose.github.io/academic/

Computer vision and assistive navigation

Software engineer with research experience at the University of Algarve's Vision Laboratory. My academic work focused on computer vision to support the mobility of blind and visually impaired people, from real-time path and obstacle detection to indoor navigation.

Research experience

Computer Vision Researcher and Research Fellow

2008 – 2013

Vision Laboratory, University of Algarve

Faro, Portugal. Developed human-vision models and active-vision algorithms in C/C++ and Matlab. Research member of ISR-Lisbon/IST, collaborating with the University of Trás-os-Montes and Alto Douro (UTAD).

SmartVision. Contributed to a portable navigation aid combining computer vision, geographic information and localisation. Detected walkable paths and static or moving obstacles beyond the reach of the white cane.

Blavigator. Worked on path following, obstacle detection and indoor landmarks, including lateral-door detection and sound feedback for navigation guidance.

Research grants and awards

SmartVision research grant

Active vision for the blind.
PTDC/EIA/73633/2006

Blavigator research grant

A cheap and reliable navigation aid for the blind.
RIPD/ADA/109690/2009

Doctoral research grant

Robust visual recognition of objects and environments for blind persons.
SFRH/BD/82310/2011

CERATONIA award grant

The Amateur Painter.
3.70.13.001

Contact and profiles

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 github.com/joaotiagojose

Google Scholar · scholar.google.com/citations?user=3z3wfQMAAAAJ

Publications

Journal articles

Computer vision and GIS for the navigation of blind persons in buildings 2015

M. Serrão, S. Shahrabadi, M. Moreno, **J. T. José**, J. I. Rodrigues, J. M. F. Rodrigues & J. M. H. du Buf
Universal Access in the Information Society, 14, 67-80. March 2015; online 19 February 2014.
DOI: 10.1007/s10209-013-0338-8

The SmartVision local navigation aid for blind and visually impaired persons 2011

J. José, M. Farrajota, J. M. F. Rodrigues & J. M. H. du Buf
International Journal of Digital Content Technology and its Applications, 5(5), 362-375.
DOI: 10.4156/jdcta.vol5.issue5.40

The SmartVision Navigation Prototype for Blind Users 2011

J. M. H. du Buf, J. Barroso, J. M. F. Rodrigues, H. Paredes, M. Farrajota, H. Fernandes, **J. José**, V. Teixeira & M. Saleiro
International Journal of Digital Content Technology and its Applications, 5(5), 351-361.
DOI: 10.4156/jdcta.vol5.issue5.39

Conference papers

Realtime Local Navigation for the Blind: Detection of Lateral Doors and Sound Interface 2012

M. Moreno, S. Shahrabadi, **J. José**, J. M. H. du Buf & J. M. F. Rodrigues
DSAI 2012. Procedia Computer Science, 14, 74-82. DOI: 10.1016/j.procs.2012.10.009

Visual Navigation for the Blind - Path and Obstacle Detection 2012

J. José, J. M. H. du Buf & J. M. F. Rodrigues
1st International Conference on Pattern Recognition Applications and Methods (ICPRAM), 515-519.
DOI: 10.5220/0003711405150519

A vision system for detecting paths and moving obstacles for the blind 2010

J. José, M. Farrajota, J. M. F. Rodrigues & J. M. H. du Buf
3rd International Conference on Software Development for Enhancing Accessibility and Fighting Info-exclusion (DSAI), Oxford, 175-182.

The SmartVision Navigation Prototype for the Blind 2010

J. M. H. du Buf, J. Barroso, J. M. F. Rodrigues, H. Paredes, M. Farrajota, H. Fernandes, **J. José**, V. Teixeira & M. Saleiro
3rd International Conference on Software Development for Enhancing Accessibility and Fighting Info-exclusion (DSAI), Oxford, 167-174.

Conference abstract

Blavigator: a navigation aid for blind persons 2012

J. Jose, M. Moreno, J. Pinilla-Dutoit, J. Rodrigues & J. Du Buf
ECVP 2012. Perception, 41(1_suppl), p. 99, abstract 97. Supplement DOI: 10.1177/03010066120410s101

Book

Real-time path and obstacle detection for blind persons 2013

João Tiago Pereira Nunes José
LAP LAMBERT Academic Publishing. ISBN 978-3-659-45730-2.

Education and teaching

Education

Doctoral studies in Computer Science Engineering

Sep 2011 – Oct 2013

University of Algarve

Completed the first two years of the programme; doctorate not completed, studies suspended. Research on robust visual recognition of objects and environments for blind users: literature review, algorithm comparison, an initial object-recognition model, and hand and arm segmentation.

MSc in Computer Science Engineering

Sep 2008 – Jun 2011

University of Algarve

Degree grade: 16/20. Dissertation grade: 18/20.

Dissertation: Real-time path and obstacle detection for blind persons

2010

C/C++, OpenCV and Qt on a netbook; supervised by Prof. Johannes Martinus Hubertina du Buf.
Repository: hdl.handle.net/10400.1/944

BSc in Computer Science Engineering

Sep 2003 – Jul 2008

University of Algarve

Selected project: a web content-management system for football news and results, using HTML, PHP, JavaScript, PostgreSQL and PL/pgSQL. Project grade: 19/20.

Studies in Electronics Engineering

Sep 2001 – Jul 2003

University of Algarve

Changed major to Computer Science Engineering; no degree awarded in Electronics Engineering.

Teaching

Cisco CCNA Exploration Instructor

Nov 2008 – Jun 2009

Cisco Academy, FCT - University of Algarve · Faro, Portugal

Network Fundamentals. Network models and protocols, Ethernet, IP addressing and subnetting.

Routing Protocols and Concepts. Routing fundamentals, VLSM, RIP, OSPF and EIGRP.

LAN Switching and Wireless. Switch configuration, VLANs, trunking, spanning tree and WLANs.

Accessing the WAN. WAN technologies, ACLs, PPP/PPPoE, Frame Relay and VPN concepts.

Training and qualifications

Professional Scrum Master I

Mar 2018

Scrum.org.

Certificate of Pedagogical Competences (CCP)

Mar 2015

IEFP. Initial trainer training in pedagogical methods and competences, Training House, Faro (Jan–Mar 2015).

Cisco Certified Network Associate (CCNA)

Oct 2007 – May 2008

Cisco Academy, FCT - University of Algarve. Modules 1-4; average grade: 90%.

Professional experience

Independent Software Engineer

Nov 2022 – Present

JTJ Software · Netherlands and Portugal

LVNL SIM Builder

Jun 2024 – Feb 2026

Full-stack development and Scrum Master responsibilities for a web-based ATM scenario platform built on an adapted BlueSky simulator.

LVNL iCAS SIM

Nov 2022 – Dec 2025

Developed and maintained the simulation and training environment for the Netherlands' next-generation air traffic management system, including peripheral-system integration.

Portuguese real-estate software

Jan 2026 – Present

Product discovery, architecture and implementation for the Portuguese real-estate market.

C++ Software Developer

May 2021 – Nov 2022

Accent Pointe B.V. · Netherlands

Developed trading software integrated with brokers and market-data feeds; modernised a legacy MFC and Microsoft RPC codebase.

C++ Software Developer

Nov 2016 – May 2021

ABN AMRO Clearing Bank · Netherlands

Delivered financial risk-management software and integrated service interfaces, messaging and authentication components.

Software Engineer

Apr 2016 – Oct 2016

Frinch · Faro, Portugal / Remote

Developed Product Information Management software with a distributed team working with the company's Munich headquarters.

Software Engineer and Lead Researcher

Oct 2013 – Sep 2015

Evision Industry Software · Faro, Portugal

Developed algorithms for text detection, raster-to-vector conversion and automatic interpretation of piping and instrumentation diagrams and AutoCAD files, including pipe and instrument recognition. Worked with the team in The Hague using C# and TypeScript.

Invited Expert

May 2007 – Aug 2007

SEF · Faro, Portugal

Participated in the quality assessment of the electronic border-control system installed at Faro Airport.

Technical skills

Research and computer vision. C/C++, Matlab, OpenCV and Qt; real-time path and obstacle detection, visual object recognition, text detection and raster-to-vector conversion.

Software engineering. C++, Python, C, Go, Ada, Bash, C# and TypeScript; HTML, PHP and JavaScript; REST APIs, ZeroMQ, Kafka and Protobuf; PostgreSQL, PL/pgSQL and Ingres; Git, CMake, GCC and GDB; pytest and C++ testing tools; Linux, Docker and OpenShift.

Languages

Portuguese · Native English · Fluent Dutch · Intermediate Spanish · Good